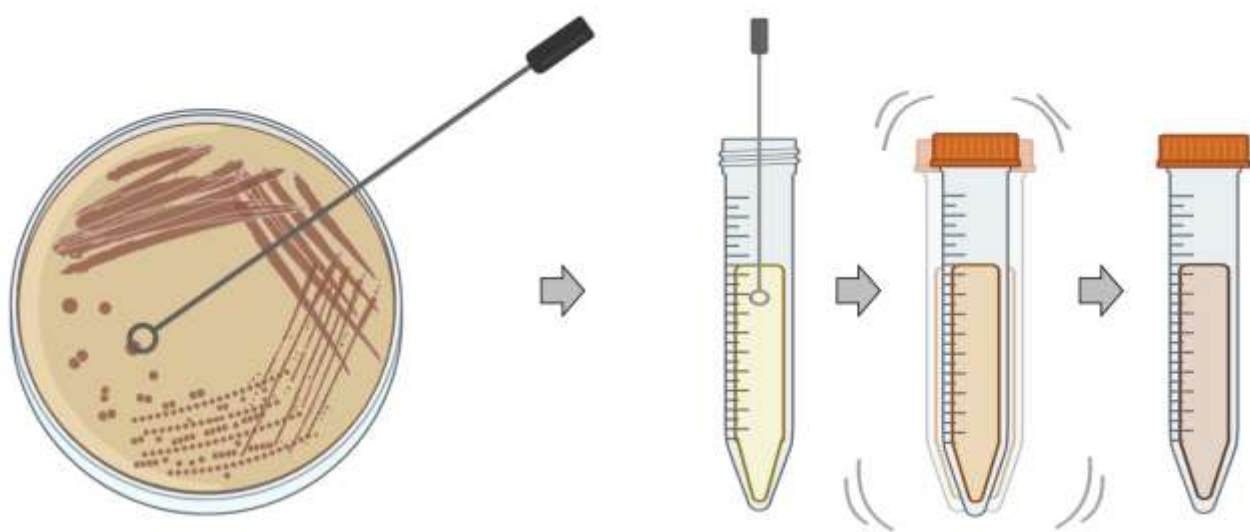


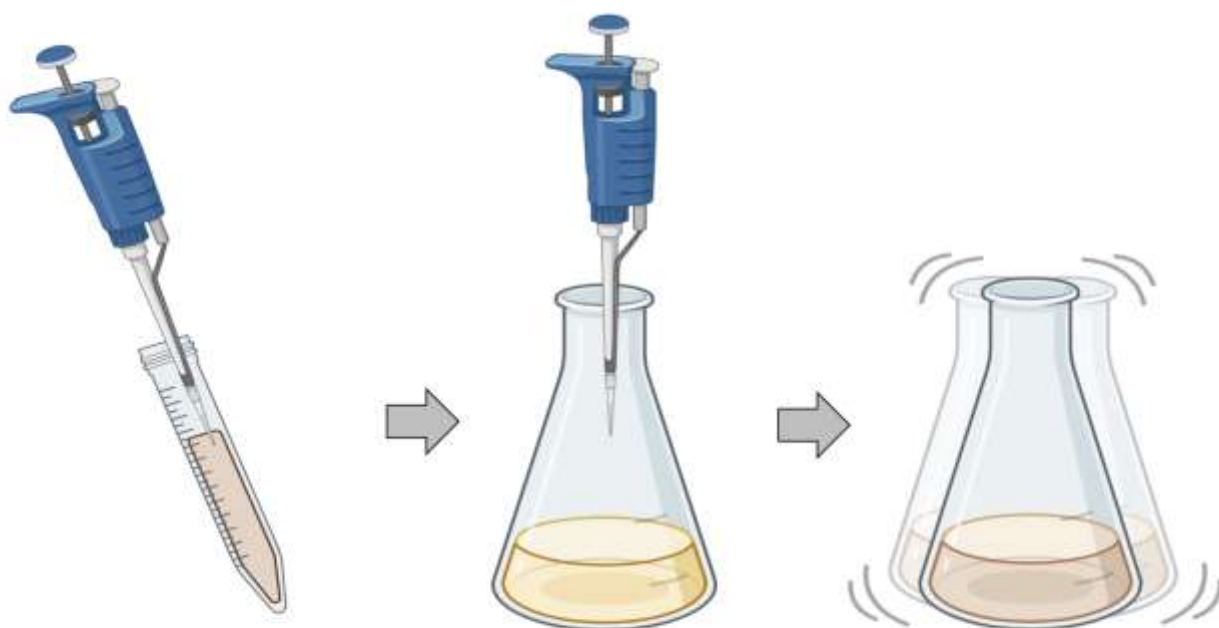
A

Preparation of *R. rhizogenes* suspension for infection

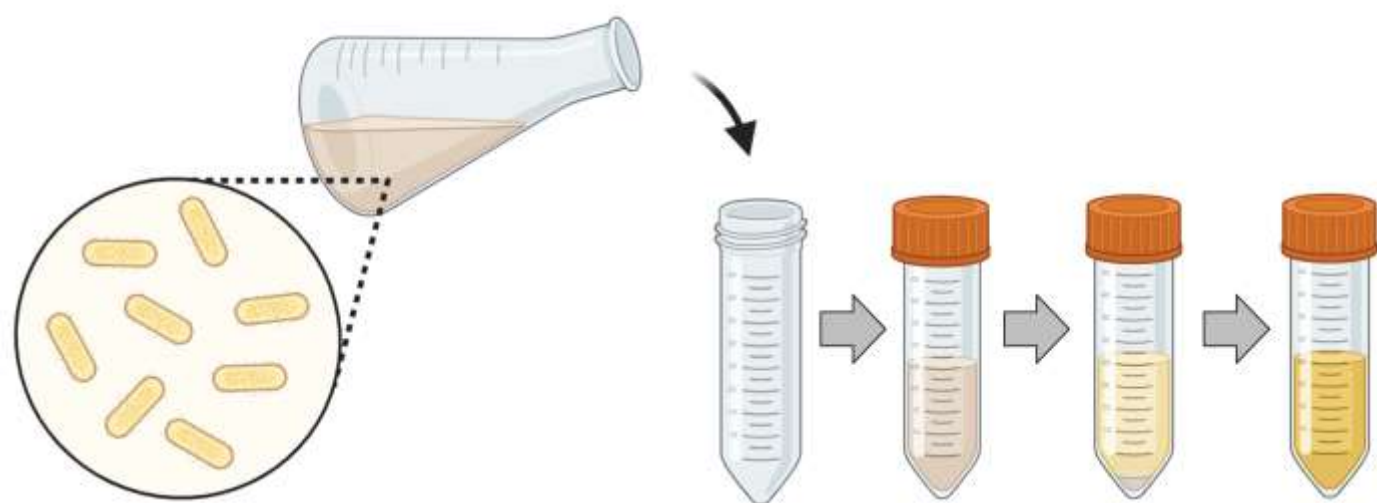


Streak out *R. rhizogenes* onto yeast mannitol (YM) agar plates supplemented with 25 $\mu\text{g}/\text{mL}$ Rifampicin and selection antibiotic (if binary vector present). Incubate plates in the dark at 28 $^{\circ}\text{C}$ for 48 h.

Prepare pre-culture by inoculating 5 mL of YM broth with a single *R. rhizogenes* colony and incubate overnight while shaking.



Use 100-200 μL of pre-culture to inoculate 50 mL of YM broth. Incubate at 28 $^{\circ}\text{C}$ in the dark while shaking at 120 rpm until $\text{OD}_{600} \approx 1$.

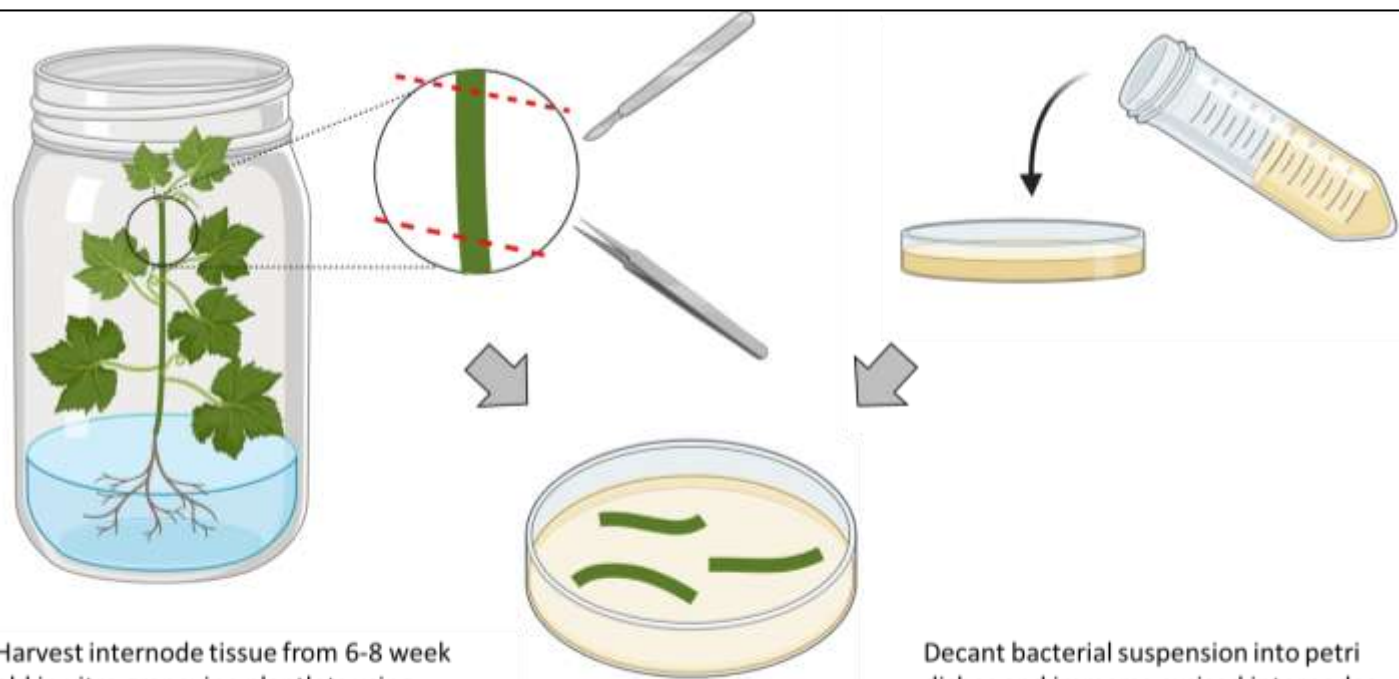


Once *R. rhizogenes* culture reaches $\text{OD}_{600} \approx 1$, decant it into falcon tubes for processing.

Centrifuge culture for 5 min at 5000 rpm, decant supernatant and wash pellet in $\frac{1}{2}$ MS. The pellet is subsequently resuspended in $\frac{1}{2}$ MS and supplemented with 250 μM acetosyringone to generate final *R. rhizogenes* culture, ready for infection.

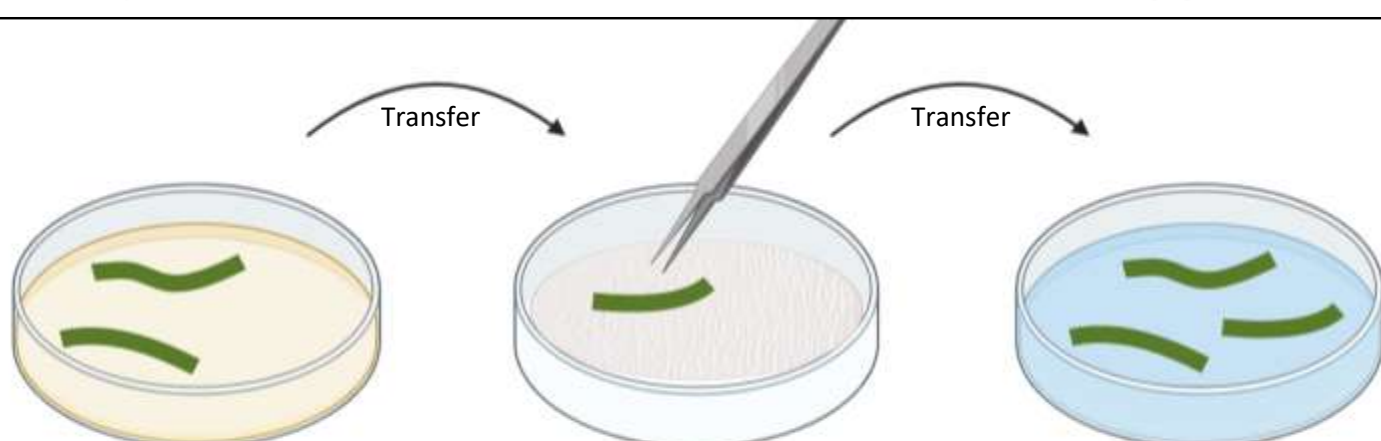
B

Infection protocol

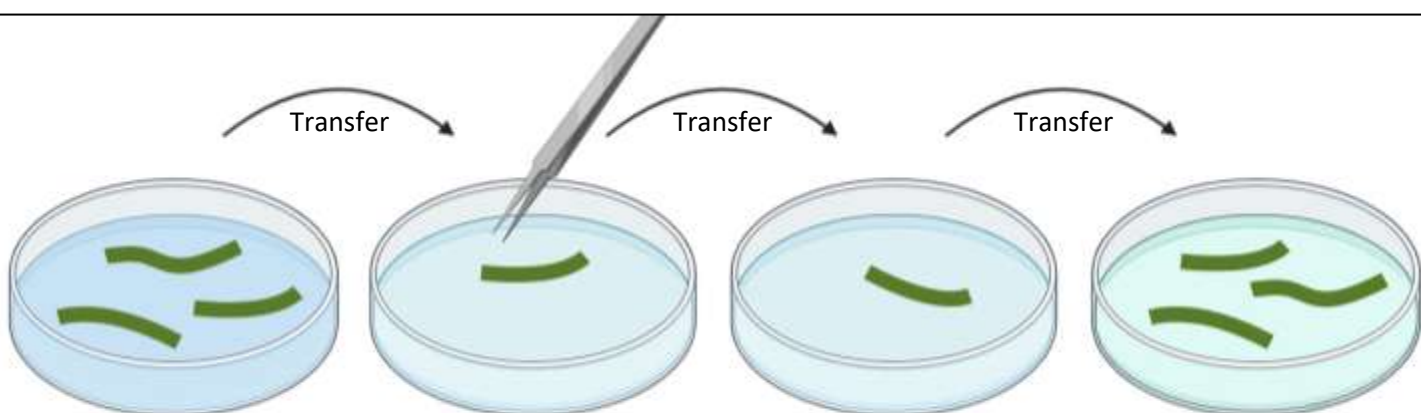


Harvest internode tissue from 6-8 week old in vitro grapevine plantlets using sterile scalpel and tweezers.

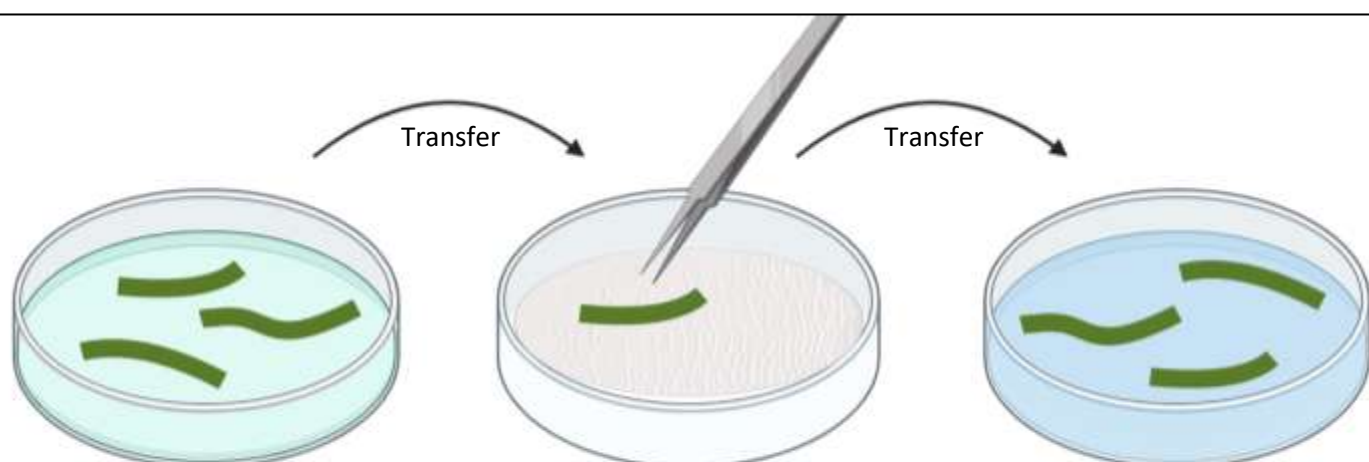
Decant bacterial suspension into petri dishes and immerse excised internodes for 20 min, avoiding agitation.



After the 20 min immersion, explants are briefly blotted dry on sterile filter paper to remove excess bacterial suspension. Subsequently, explant material and *R. rhizogenes* are co-cultured on $\frac{1}{2}$ MS agar plates for 48 h in the dark at 20 °C.

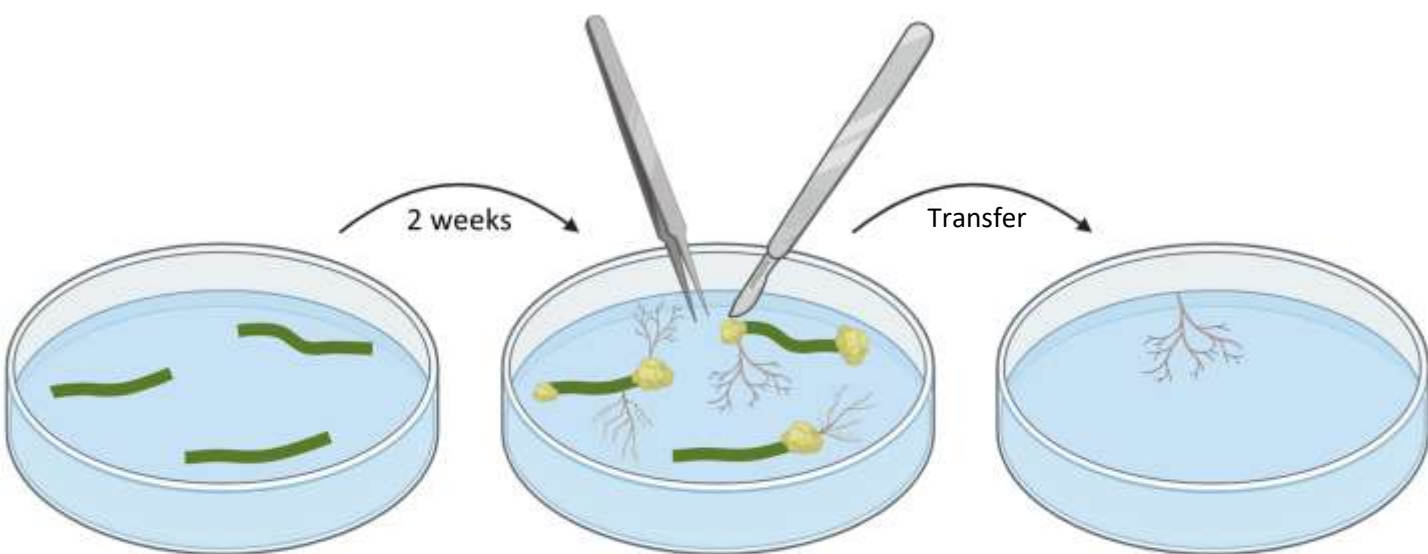


After the 48 h co-cultivation *R. rhizogenes* is eliminated. Explants are rinsed twice in distilled water followed by submersion in an antibiotic solution (500 $\mu\text{g}/\text{mL}$ Timentin) for 30 min to remove the bacterium.

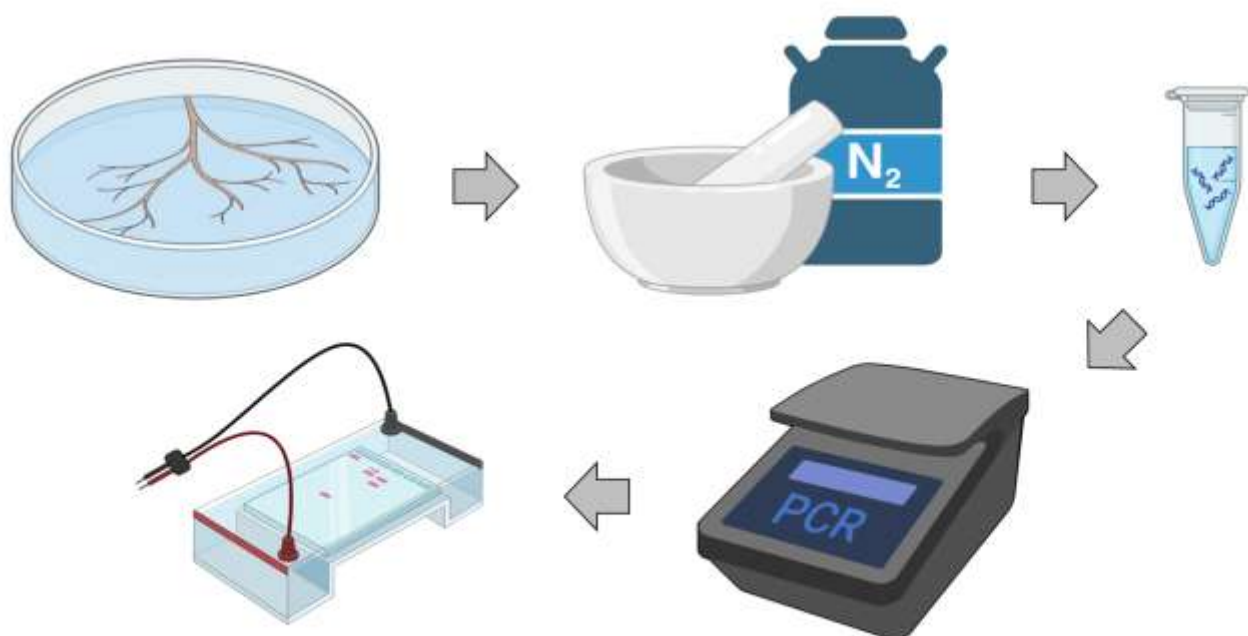


Thereafter, the washed explants are blotted dry on sterile filter paper and transferred to $\frac{1}{2}$ MS plates supplemented with 200 $\mu\text{g}/\text{mL}$ Timentin. The explants are then incubated in the dark at 25 °C.

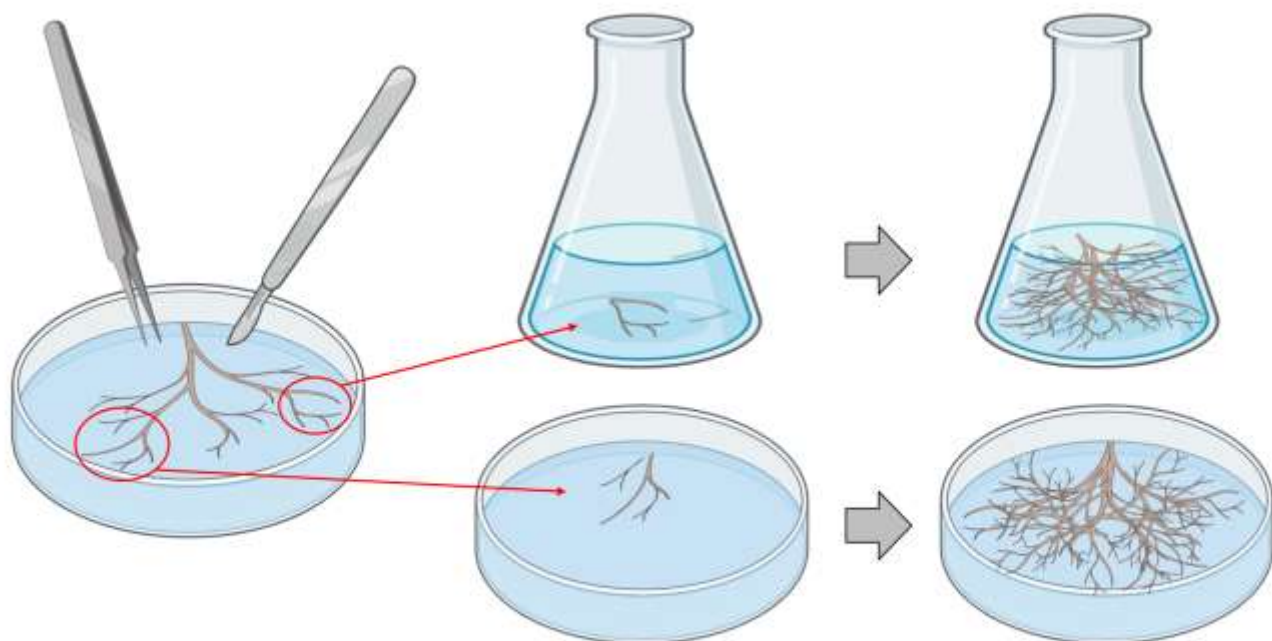
Hairy root confirmation and culturing



Roots and calli start emerging about 2 weeks after the elimination of *R. rhizogenes*. When roots reach 2-4 cm in length they are excised and transferred to $\frac{1}{2}$ MS media and grown for 4 weeks. Roots that continue growing non-geotropically and start branching when separated from its explants are considered putative hairy roots.



Putative hairy root cultures are screened for transgenes to confirm hairy root status. Roots are initially flash frozen, pulverized and DNA extracted. The DNA extract is then used in PCR reactions to screen for *rol* and *vir* genes to determine if *R. rhizogenes* transformation was successful and if bacterium was eliminated.



Confirmed hairy roots can be sub-cultured by excising actively growing root tips and placing them on or in fresh $\frac{1}{2}$ MS media. This process can be repeated indefinitely.